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4.5 Energy Changes

AQA GCSE Chemistry (8462) — Paper 1

Every chemical reaction involves an energy change. Bonds in the reactants must be **broken** and new bonds in the products must be **made**, and the balance between these two processes decides whether a reaction gives out energy to the surroundings or takes it in. This topic covers how we recognise and measure these energy changes, how we show them on **reaction profiles**, how we calculate them from **bond energies** (HT), and how chemical reactions are used to generate electricity in **cells, batteries and fuel cells** (Triple).

Content marked **(HT)** is Higher Tier only. Content marked **(Triple)** is only examined in GCSE Chemistry (8462), not in Combined Science.

4.5.1 Exothermic and Endothermic Reactions

4.5.1.1 Energy Transfer During Exothermic and Endothermic Reactions

> Energy is **conserved** in chemical reactions. The amount of energy in the universe at the end of a chemical reaction is the same as before the reaction takes place. If a reaction transfers energy **to the surroundings**, the product molecules must have **less** energy than the reactants, by the amount transferred.

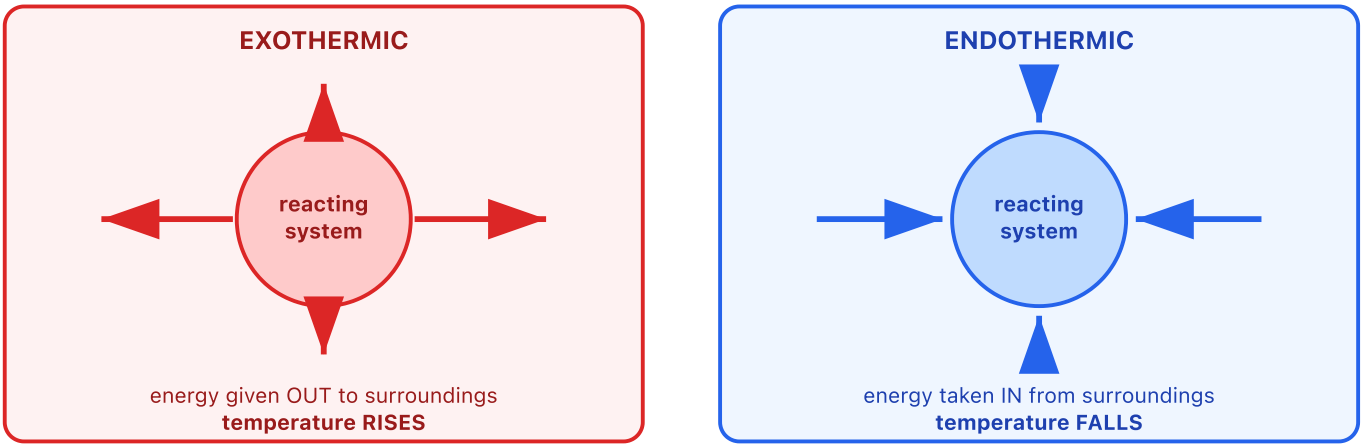
To talk about energy changes clearly we separate the **system** (the reacting chemicals) from the **surroundings** (everything else — the solvent water, the test tube, the thermometer, the air). A thermometer dipped into a reaction mixture measures the temperature of the surroundings, so it tells us which way energy is flowing.

> An **exothermic reaction** is one that **transfers energy to the surroundings**, so the temperature of the surroundings **increases**.

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An **endothermic reaction** is one that **takes in energy from the surroundings**, so the temperature of the surroundings **decreases**.

Direction of energy transfer



Examples of Exothermic and Endothermic Reactions

Exothermic (temperature rises)	Endothermic (temperature falls)
Combustion — burning fuels, e.g. $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$	**Thermal decomposition**, e.g. $CaCO_3 \rightarrow CaO + CO_2$ (needs continuous heating)
Many oxidation reactions, e.g. rusting, metals reacting with oxygen; respiration	The reaction of **citric acid and sodium hydrogencarbonate** (e.g. sherbet fizzing on the tongue feels cold)
Neutralisation, e.g. $HCl + NaOH \rightarrow NaCl + H_2O$	**Photosynthesis** (energy taken in from light)
Displacement, e.g. $Mg + CuSO_4 \rightarrow MgSO_4 + Cu$	Dissolving **ammonium nitrate** in water (a physical change, but endothermic)
Metals reacting with acids, e.g. $Mg + 2HCl \rightarrow MgCl_2 + H_2$	Electrolysis (energy supplied electrically)

Everyday Uses

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Exothermic and endothermic reactions have everyday uses that you should be able to evaluate:

Product	Type	How it works	Evaluation
Self-heating cans (coffee, soup)	Exothermic	Pressing a button mixes water with **calcium oxide** in a separate compartment: $CaO + H_2O \rightarrow Ca(OH)_2$. The energy released heats the food or drink.	Convenient where there is no cooker (camping, military). But expensive, bulky, single-use, and the heat output is limited — contents may not heat evenly.
Disposable hand warmers	Exothermic	**Iron powder** is oxidised by air (with salt and water as catalysts) — essentially fast rusting. Lasts several hours.	Long-lasting, cheap; but cannot be reused.
Reusable hand warmers	Exothermic	A supersaturated solution of **sodium ethanoate** crystallises when a metal disc is clicked, releasing energy. Boiling the pack re-dissolves the crystals.	Reusable many times, but heat lasts a shorter time and the pack must be reset in boiling water.
Instant cold packs (sports injuries)	Endothermic	Squeezing the pack breaks an inner bag, so water dissolves **ammonium nitrate** (or reacts with other chemicals), taking in energy and cooling the injury.	Instantly available without a freezer; reduces swelling. Single use only; chemicals may be harmful if the pack splits.

Required Practical 4 — Temperature Changes

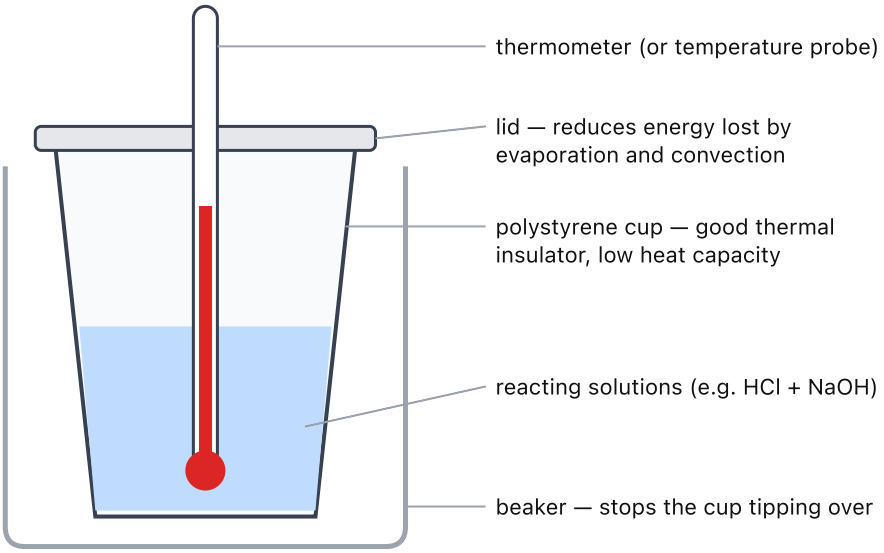
Aim: to investigate the variables that affect temperature changes in reacting solutions, such as acid plus metals, acid plus carbonates, neutralisations, and displacement of metals.

A typical version: investigate how the **volume of sodium hydroxide** added to a fixed volume of hydrochloric acid affects the temperature change.

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1. Using a measuring cylinder, put **30 cm³ of dilute hydrochloric acid** into a **polystyrene cup**. Stand the cup inside a **beaker** so it does not fall over.
2. Measure and record the **initial temperature** of the acid with a thermometer.
3. Measure **5 cm³ of sodium hydroxide solution** and add it to the cup. Fit a **lid** with a hole for the thermometer.
4. **Stir** the solution with the thermometer and record the **highest temperature** reached.
5. Repeat, adding further 5 cm³ portions of NaOH, recording the highest temperature after each addition, up to about 40 cm³ in total.
6. Repeat the whole experiment to get a second set of results and calculate mean temperatures. Plot a graph of **highest temperature** against **volume of NaOH added**.

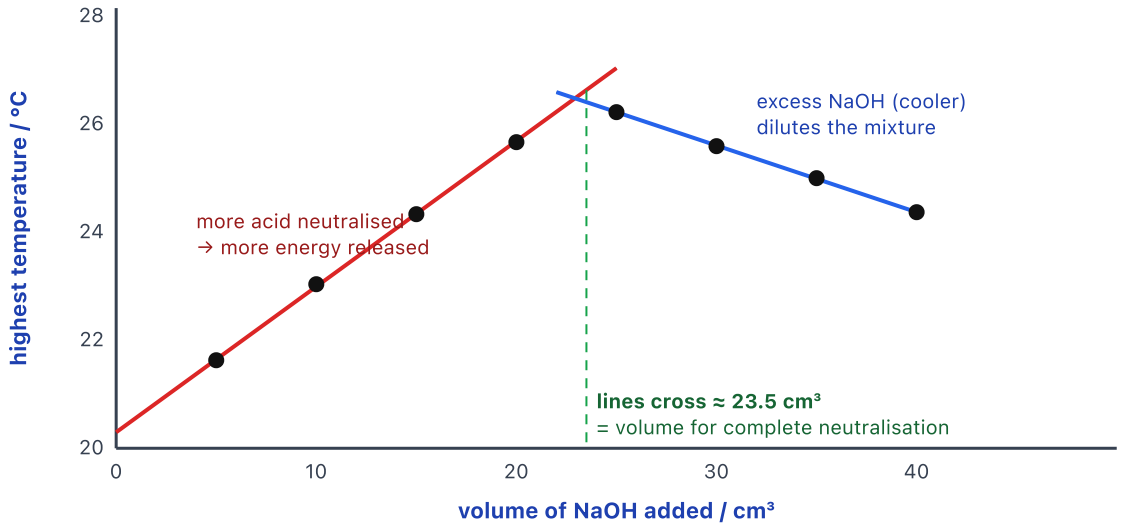
Measuring temperature change in a reaction



Variable	In this investigation
Independent	Volume of sodium hydroxide added (or: type of metal, concentration of acid, mass of solid…)
Dependent	Highest temperature reached / temperature change
Control	Volume and concentration of acid; concentration of alkali; same cup and lid; same starting temperature; stirring

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Typical results — highest temperature vs volume of NaOH



The temperature rises as more acid is neutralised. Once all the acid has reacted, adding more (cooler) alkali just **dilutes** the mixture and absorbs energy, so the temperature falls. Drawing two straight **lines of best fit** and finding where they cross gives the volume of alkali needed for exact neutralisation and the true maximum temperature.

> **Exam Tip:** The main source of error in this practical is **energy lost to the surroundings**, which makes measured temperature changes smaller than the true value. Improvements to suggest: use a lid, use a polystyrene cup (or add more insulation), stir so the temperature is even, and read the highest temperature quickly. A temperature probe and data logger reduces reading errors. A polystyrene cup is used rather than a glass beaker because polystyrene is a **poor conductor** of heat.

Worked Example 1 — Interpreting Temperature Data

Question: A student added 0.5 g of four different metal powders to 25 cm³ of copper(II) sulfate solution and recorded the temperatures. Deduce the order of reactivity.

Metal	Start / °C	Highest / °C	Change / °C
Magnesium	20.5	58.0	+37.5
Zinc	20.5	35.5	+15.0
Iron	20.0	27.0	+7.0
Silver	20.5	20.5	0.0

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```
Working: temperature change = highest − start
         The bigger the temperature rise, the more energy is
         released, so the bigger the difference in reactivity
         between the metal and copper.

Answer:  Mg > Zn > Fe > Cu > Ag
         (silver gives no rise: it is less reactive than copper,
          so no displacement occurs)
```

Possible RP4 investigation	Independent variable	Control variables
Acid + metal	Type of metal (or mass of metal, or concentration of acid)	Volume and concentration of acid; mass and surface area of metal
Acid + carbonate	Mass of carbonate (e.g. sodium hydrogencarbonate) — temperature *falls*	Volume and concentration of acid; particle size
Neutralisation	Volume or concentration of alkali	Volume and concentration of acid; starting temperature
Displacement	Type of metal added to copper(II) sulfate	Mass of metal powder; volume and concentration of solution

4.5.1.2 Reaction Profiles

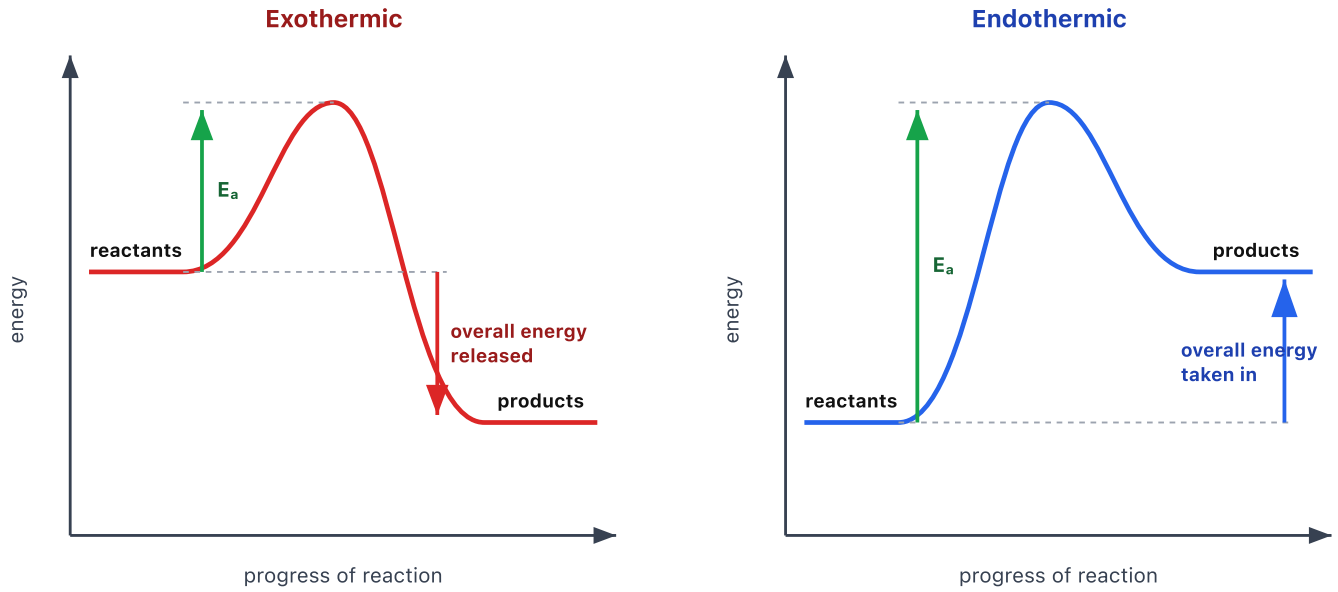
> Chemical reactions can only occur when reacting particles **collide** with each other and with **sufficient energy**. The minimum amount of energy that particles must have to react is called the **activation energy** (E_a).

Reaction profiles (energy level diagrams) can be used to show the relative energies of reactants and products, the activation energy and the overall energy change of a reaction. You must be able to draw simple reaction profiles for exothermic and endothermic reactions, showing the relative

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energies of reactants and products, the activation energy and the overall energy change, with a curved line to show the energy as the reaction proceeds.

Reaction profiles



Feature	Exothermic	Endothermic
Products vs reactants	Products **lower** in energy than reactants	Products **higher** in energy than reactants
Overall energy change arrow	Points **down** (energy released)	Points **up** (energy absorbed)
Sign of energy change, ΔH	**Negative** (e.g. −890 kJ/mol)	**Positive** (e.g. +178 kJ/mol)
Temperature of surroundings	Increases	Decreases
Activation energy	Always measured from the **reactants** up to the **peak** of the curve. Both types need activation energy.	

Exam Tip: The two most common mistakes on reaction profiles are (1) drawing the activation energy arrow from the *products* to the peak, or from the axis — it must go from

the **reactant level** to the **top of the hump**; and (2) forgetting arrowheads, or putting them on both ends. Label reactants and products, and label each arrow. A catalyst lowers the hump (see 4.6.1.4) but does not change the reactant or product levels.

4.5.1.3 The Energy Change of Reactions (HT)

During a chemical reaction:

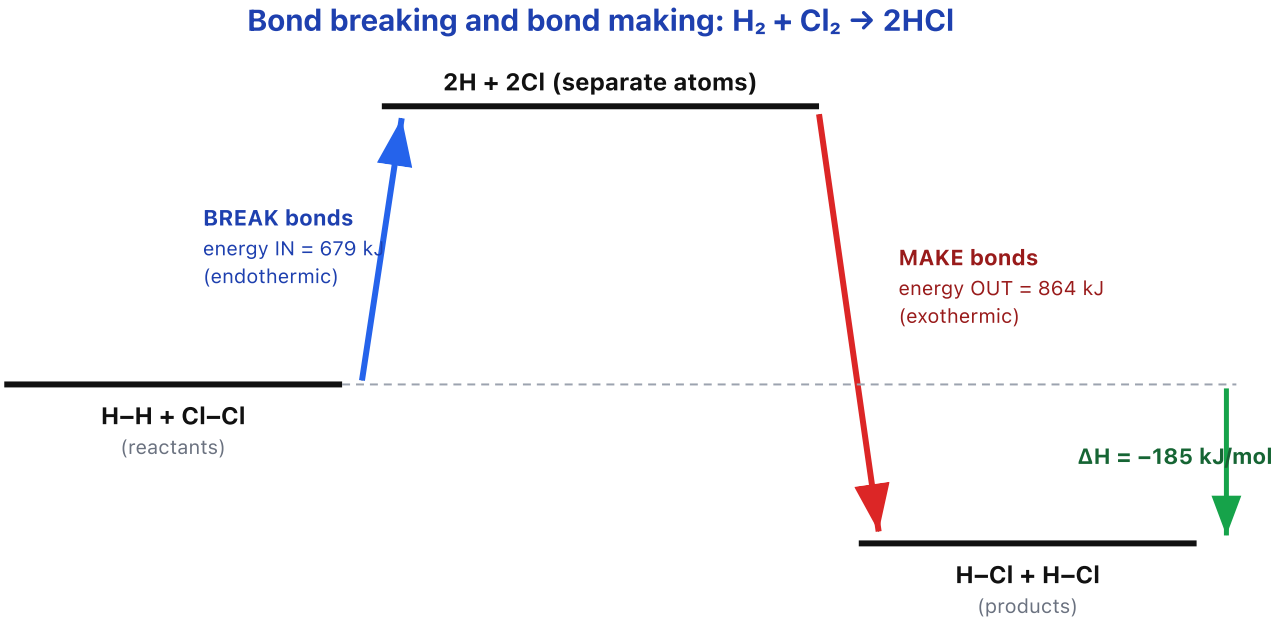
- **Energy must be supplied to break bonds** in the reactants — bond breaking is **endothermic**.
- **Energy is released when bonds are formed** in the products — bond making is **exothermic**.

The energy needed to break bonds and the energy released when bonds are formed can be calculated from **bond energies** (the energy needed to break one mole of a particular bond, in kJ/mol).

```
overall energy change = energy needed to break bonds in reactants – energy released making bonds in products
```

In an **exothermic** reaction, the energy released from forming new bonds is **greater** than the energy needed to break existing bonds → ΔH is **negative**.
In an **endothermic** reaction, the energy needed to break existing bonds is **greater** than the energy released from forming new bonds → ΔH is **positive**.

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More energy released making bonds than taken in breaking them → EXOTHERMIC

Typical Bond Energies

Values are always given in the question — you do not need to memorise them.

Bond	Bond energy (kJ/mol)	Bond	Bond energy (kJ/mol)
H–H	436	C–H	413
Cl–Cl	243	C–C	347
Br–Br	193	C=C	612
H–Cl	432	C=O	805
H–Br	366	O=O	498
O–H	464	N≡N	945
N–H	391	C–Cl	346

Worked Example 2 — Hydrogen and Chlorine (HT)

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```
Question: Calculate the energy change for:  H₂ + Cl₂ → 2HCl
          (H–H = 436, Cl–Cl = 243, H–Cl = 432 kJ/mol)

Bonds broken (reactants):  1 × H–H   = 436
                           1 × Cl–Cl = 243
                           Total     = 679 kJ

Bonds made (products):     2 × H–Cl  = 2 × 432 = 864 kJ

Energy change = broken – made
              = 679 – 864
              = –185 kJ/mol

Answer: –185 kJ/mol. Negative, so the reaction is EXOTHERMIC.
```

Worked Example 3 — Combustion of Methane (HT)

```
Question: Calculate the energy change for the complete combustion
          of methane:  CH₄ + 2O₂ → CO₂ + 2H₂O

  Displayed:   H                       O=C=O   +  2 × H–O–H
               |
           H–C–H   +  2 × O=O
               |
               H

Bonds broken:  4 × C–H = 4 × 413 = 1652
               2 × O=O = 2 × 498 =  996
               Total             = 2648 kJ

Bonds made:    2 × C=O = 2 × 805 = 1610
               4 × O–H = 4 × 464 = 1856
               Total             = 3466 kJ

Energy change = 2648 – 3466 = –818 kJ/mol   (exothermic)
```

Worked Example 4 — An Endothermic Reaction (HT)

```
Question: Calculate the energy change for:  2HBr → H₂ + Br₂
          (H–Br = 366, H–H = 436, Br–Br = 193 kJ/mol)
```

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```
Bonds broken:  2 × H–Br = 2 × 366 = 732 kJ
Bonds made:    1 × H–H  = 436
               1 × Br–Br = 193
               Total    = 629 kJ

Energy change = 732 − 629 = +103 kJ/mol

Answer: +103 kJ/mol. Positive, so the reaction is ENDOTHERMIC.
```

Worked Example 5 — Finding an Unknown Bond Energy (HT)

```
Question: For  N₂ + 3H₂ → 2NH₃   the energy change is −93 kJ/mol.
          N≡N = 945 kJ/mol, H–H = 436 kJ/mol.
          Calculate the N–H bond energy.

Bonds broken:  1 × N≡N = 945
               3 × H–H = 1308
               Total   = 2253 kJ

Bonds made:    2 NH₃ contain 6 × N–H = 6x

Energy change = broken − made
        −93   = 2253 − 6x
         6x   = 2253 + 93 = 2346
          x   = 391 kJ/mol

Answer: N–H = 391 kJ/mol
```

Exam Tip: Draw out the **displayed formula** of every molecule before counting bonds — it is very easy to miss that CO_2 has *two* C=O double bonds, or that $2H_2O$ contains *four* O–H bonds. Remember to multiply by the balancing numbers. Always write "**bonds broken – bonds made**" and keep the sign in your answer: a positive value without a sign may be marked wrong. Bond-energy answers often differ slightly from measured values because bond energies are **averages** taken across many different compounds.

Common Misconceptions

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Misconception	Correct idea
"Breaking bonds releases energy."	Breaking bonds always **requires** energy (endothermic). Energy is released only when new bonds **form**.
"An endothermic reaction feels hot because it takes in heat."	It takes energy **from the surroundings**, so the surroundings (including your hand) get **colder**.
"Exothermic reactions do not need activation energy."	All reactions need activation energy. Burning methane needs a spark or flame to start; once started, the energy released keeps it going.
"A catalyst changes the overall energy change."	A catalyst lowers the **activation energy** only; the reactant and product energy levels, and therefore ΔH, are unchanged.
"Bond energy calculations give exact answers."	Bond energies are **mean values** averaged over many compounds, so calculated energy changes are approximate.

4.5.2 Chemical Cells and Fuel Cells (Triple)

4.5.2.1 Cells and Batteries

> A **cell** contains chemicals which react to produce electricity. A simple cell can be made by connecting **two different metals** in contact with an **electrolyte**.
>
> A **battery** consists of **two or more cells connected together in series** to provide a greater voltage.

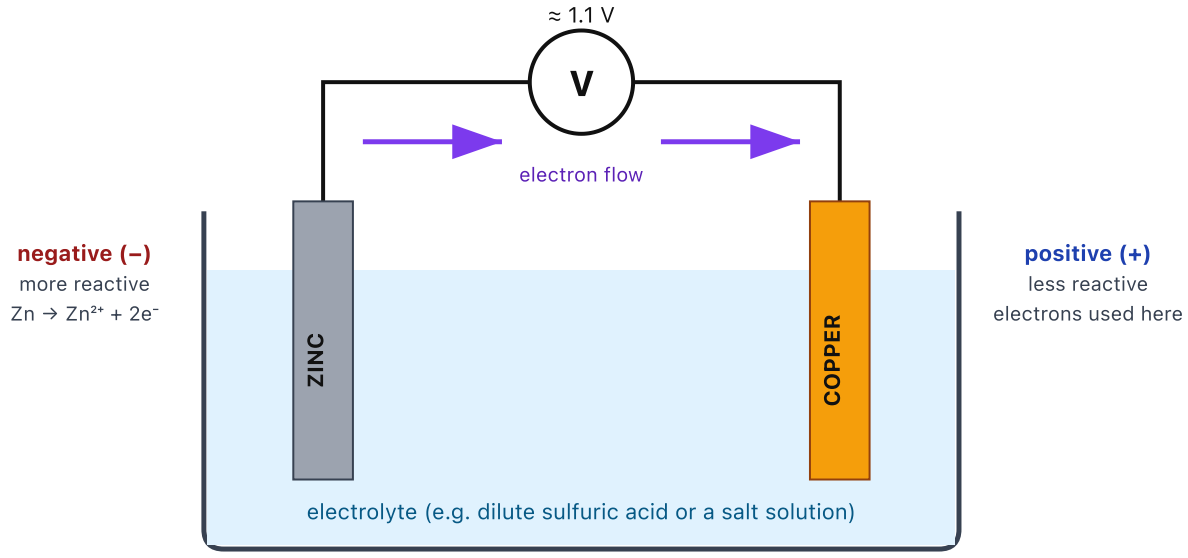
The **voltage** produced by a cell is dependent upon a number of factors, including:

- the **type of electrode** (which two metals are used) — the **bigger the difference in reactivity** between the two metals, the **larger the voltage**;
- the **electrolyte** used (its type and concentration).

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In a simple cell, the **more reactive metal** loses electrons more readily and becomes the **negative** terminal. Electrons flow through the external circuit from the more reactive metal to the less reactive metal. If both electrodes are the **same metal**, the voltage is **zero**.

A simple chemical cell (zinc and copper)



Interpreting Cell Data

You should be able to interpret data for **relative reactivity** of different metals and evaluate the use of cells. The table below shows voltages measured with a copper electrode and a second metal in the same electrolyte:

Metal paired with copper	Voltage / V	Deduction
Magnesium	2.7	Largest difference in reactivity — magnesium is the most reactive
Zinc	1.1	Zinc is less reactive than magnesium
Iron	0.8	Iron is less reactive than zinc
Lead	0.5	Lead is just above copper
Copper	0.0	Same metals — no difference in reactivity, no voltage

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Order of reactivity deduced: **Mg > Zn > Fe > Pb > Cu**. If a metal less reactive than copper (e.g. silver) were used, the voltage would have the *opposite* sign because copper would then become the negative terminal.

Rechargeable and Non-Rechargeable Cells

Type	Explanation	Examples
Non-rechargeable	The chemical reactions stop when one of the **reactants has been used up**. The cell is then "flat" and must be replaced.	**Alkaline batteries** (zinc and manganese dioxide), zinc–carbon cells
Rechargeable	The chemical reactions are **reversed when an external electrical current is supplied**, regenerating the original reactants.	Lithium-ion (phones, laptops, electric cars), nickel–metal hydride, lead–acid car batteries

> **Exam Tip:** If asked why a non-rechargeable battery eventually stops working, say "one of the **reactants is used up**, so the chemical reaction stops". If asked why a rechargeable battery can be recharged, say "the **reaction is reversed** when an external current is supplied". Vague answers like "it runs out of energy" or "the electricity is used up" do not score.

4.5.2.2 Fuel Cells

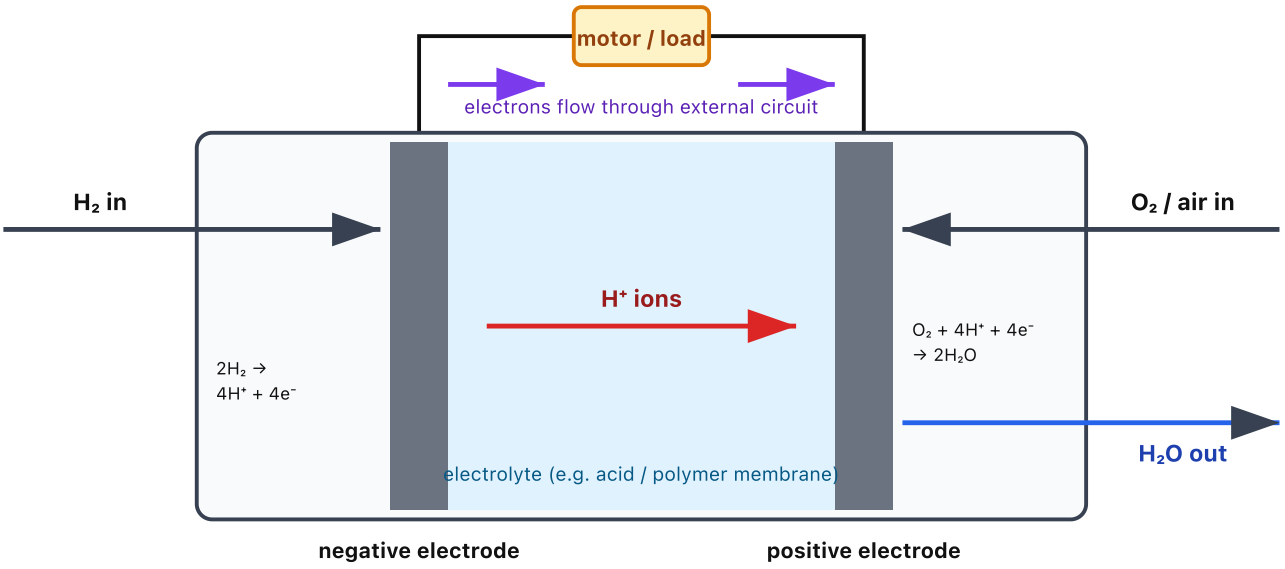
> **Fuel cells** are supplied by an **external source of fuel** (e.g. hydrogen) and **oxygen or air**. The fuel is **oxidised electrochemically** within the fuel cell to produce a potential difference. Unlike a battery, a fuel cell keeps producing electricity for as long as fuel and oxygen are supplied.

The overall reaction in a **hydrogen fuel cell** involves the oxidation of hydrogen to produce **water**:

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$$

> This is the same overall reaction as burning hydrogen, but in a fuel cell the energy is released as **electrical energy** rather than mostly as heat, and there is no flame.

Hydrogen–oxygen fuel cell (acidic electrolyte)



Half Equations in the Hydrogen Fuel Cell (HT)

Acidic electrolyte (the version used by AQA):

Negative electrode:	$2H_2 \rightarrow 4H^+ + 4e^-$	hydrogen is OXIDISED (loses electrons)
Positive electrode:	$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$	oxygen is REDUCED (gains electrons)

Add them ($4e^-$ and $4H^+$ cancel):
Overall: $2H_2 + O_2 \rightarrow 2H_2O$

Alkaline electrolyte (for interest — e.g. KOH fuel cells):

Negative electrode:	$2H_2 + 4OH^- \rightarrow 4H_2O + 4e^-$	
Positive electrode:	$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$	
Overall:	$2H_2 + O_2 \rightarrow 2H_2O$	(same)

Evaluating Hydrogen Fuel Cells vs Rechargeable Batteries

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Hydrogen fuel cells — advantages	Hydrogen fuel cells — disadvantages
• The **only product is water** — no CO_2 or other pollutants at the point of use. • Do not need to be electrically **recharged**; refuelling takes minutes rather than hours. • Keep working **as long as fuel is supplied** — longer range for vehicles. • Fewer parts to wear out; do not lose efficiency over repeated charge cycles. • Less toxic waste at the end of life than batteries (which contain toxic metal compounds).	• **Hydrogen is a gas** — it takes up a large volume and must be stored under **high pressure** or at very low temperature. • Hydrogen is **highly flammable** — risk of explosion if it leaks. • Hydrogen is usually made from **fossil fuels** (e.g. methane) or by electrolysis using electricity that may come from fossil fuels — so it is not necessarily carbon-free overall. • Few hydrogen **refuelling stations**; fuel cells are expensive (platinum catalysts).

Rechargeable batteries — advantages	Rechargeable batteries — disadvantages
• Can be recharged from the mains anywhere there is a socket. • No flammable gas storage needed. • Well-developed technology; widely available.	• Recharging takes a long time. • Store less energy per kg — limited range in vehicles. • Lose capacity after many charge cycles; contain toxic/scarce metals that are hard to dispose of. • Electricity used for charging may be generated from fossil fuels.

Exam Tip: Evaluation questions ("Evaluate the use of hydrogen fuel cells compared with rechargeable cells for powering vehicles") need points on **both sides** and a **justified conclusion**. The key chemistry point is that a fuel cell's only product is **water**, but the

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> hydrogen itself may have been produced using fossil fuels. Do not say that fuel cells are "100% clean" without qualifying it.

Summary Checklist for 4.5 Energy Changes

- State that energy is conserved in chemical reactions.
- Define exothermic and endothermic reactions in terms of energy transfer to or from the surroundings and temperature change.
- Give examples of each: combustion, oxidation, neutralisation (exo); thermal decomposition, citric acid + sodium hydrogencarbonate (endo).
- Describe and evaluate everyday uses: self-heating cans, hand warmers, sports injury cold packs.
- Describe Required Practical 4 — measuring temperature changes in a polystyrene cup — and identify variables and sources of error.
- Define activation energy; draw and label reaction profiles for exothermic and endothermic reactions.
- (HT) Explain that bond breaking is endothermic and bond making is exothermic.
- (HT) Calculate the energy change of a reaction from bond energies: `bonds broken − bonds made`, including finding an unknown bond energy.
- (Triple) Describe a simple cell; explain how the voltage depends on the electrodes and electrolyte; interpret cell data to deduce reactivity.
- (Triple) Explain why non-rechargeable cells stop working and how rechargeable cells are recharged.
- (Triple) Describe a hydrogen fuel cell; write the overall equation; (HT) write half equations for each electrode.
- (Triple) Evaluate hydrogen fuel cells against rechargeable batteries.

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